

BZ 214 Visual Programming

Traffic Light Control System Based on Vehicle Density

1. Overview

Traffic congestion is a common issue in modern cities, often made worse by static (fixed timers) traffic lights that don't respond to real-time traffic. This project asks you to create a smart traffic light system that adjusts signal times based on the number of vehicles.

You will also simulate cars moving through the intersection and make sure they do not crash into each other.

You'll develop the application in Java, using JavaFX for the graphical interface. Make sure to use object-oriented programming principles and follow the Model-View-Controller (MVC) design pattern.

2. Project Goals

- Simulate a four-way traffic intersection with vehicles moving in four directions: **North, South, East, and West**.
- Use either manually entered or randomly generated vehicle density data.
- Calculate and assign **green light durations proportionally** based on real-time density.
- Visually represent the intersection and simulate real-time traffic lights with working timers and moving cars.
- Demonstrate an understanding of JavaFX, OOP, MVC, and data structures in Java.

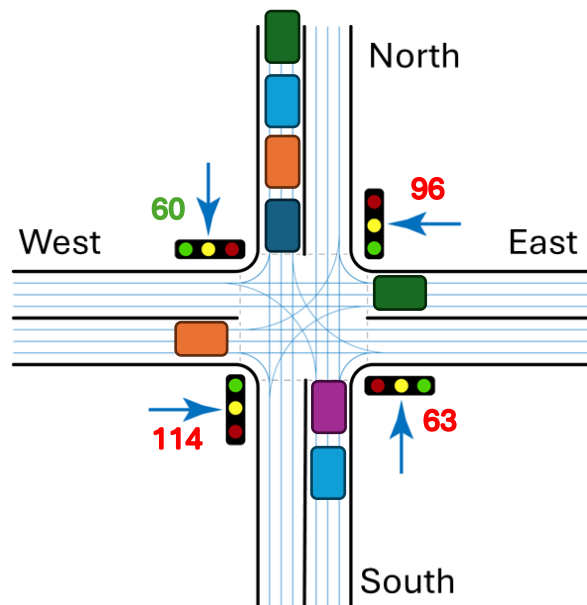


Figure 1. A sample intersection

3. Functional Requirements

Your system must support the following features:

- A **graphical user interface (GUI)** displaying the intersection, traffic lights and cars.
- Input fields or a generator for vehicle density in each direction are needed. It should allow both manually entered and randomly generated vehicle density data.
- Dynamic calculation and automatic update of signal durations.
- Display of the remaining time for green, yellow, and red phases.
- Simulation mode with **animated or timed signal transitions**.
- Simulation of animated cars that:
 - Wait when the light is red.
 - Move forward when the light turns green.
 - Disappear after passing the intersection.
 - Do not crash to each other
- No new cars enter the system during a cycle. You start with a set number of cars, and the system manages them until the cycle ends.

4. Technical Specifications

- Use **JavaFX** as GUI library.
- Strict adherence to the **MVC** architecture.
- Follow best practices in **Object-Oriented Programming**.
- Use **Java Collection Frameworks** (e.g., List, Map) as appropriate.
- **No third-party libraries** may be used—only standard **Java SE libraries** are permitted. (If you really need to use a third-party library, please ask for permission first.)

5. Signal Timing Logic

- Total cycle time: **Fixed at 120 seconds**. (Do not change the cycle time)
- Each direction's green light time is proportional to the traffic density.
- The yellow light duration is **constant**, e.g., **3 seconds**. (You can select a proper yellow light duration)

- Green light durations must fall within a **defined range** (e.g., 10 to 60 seconds).

Sample Calculation:

Direction	Vehicle Count	Percentage	Green Light Duration
North	40	50%	60 seconds
South	20	25%	30 seconds
East	10	12.5%	15 seconds
West	10	12.5%	15 seconds

6. GUI Expectations

Your application should include:

- A central graphical layout of a **four-way intersection**.
- **Input panel** for vehicle data entry or random generation.
- **Simulation controls**: “Start”, “Pause”, and “Reset” buttons.
- Digital timers showing **countdown for each light**.
- Animation of vehicle movement or signal transitions.

7. Documentation

At the end of your project, you must prepare a detailed report that describes your work. You must give details about your project, provide the class, and use case diagrams. You can find a template for your project report on the ERUDM.

8. Academic Honesty

Honesty and integrity are essential. Never submit the work of others as your own. A student who copies the code from an outside source or another one or allows someone to copy from their project will receive 0 points.

9. Submission Checklist

Please submit a **ZIP archive** named with your **group number (e.g. group1.zip)**, containing the following:

- All the source code files (.java) required to compile and run your program
- A usage manual or README
- Project report

- (Optional) Simulation video or animated GIF

If you have any questions or concerns about the submission process, please don't hesitate to ask.

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